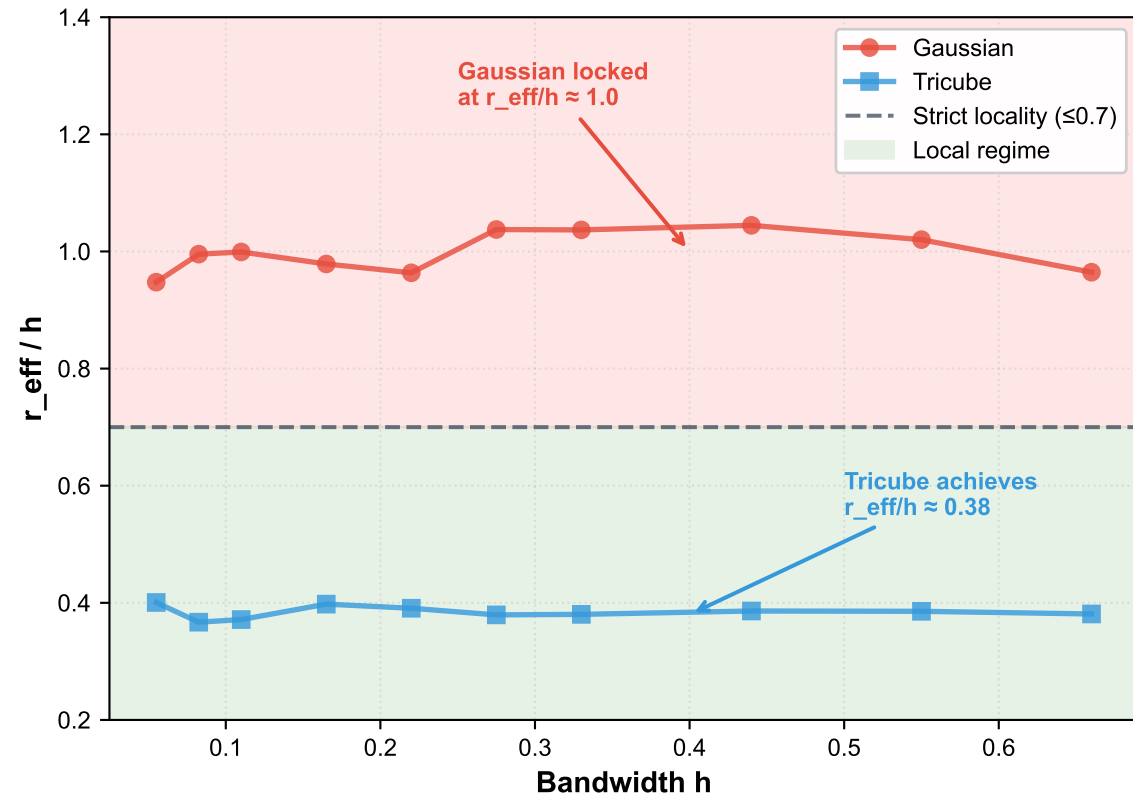
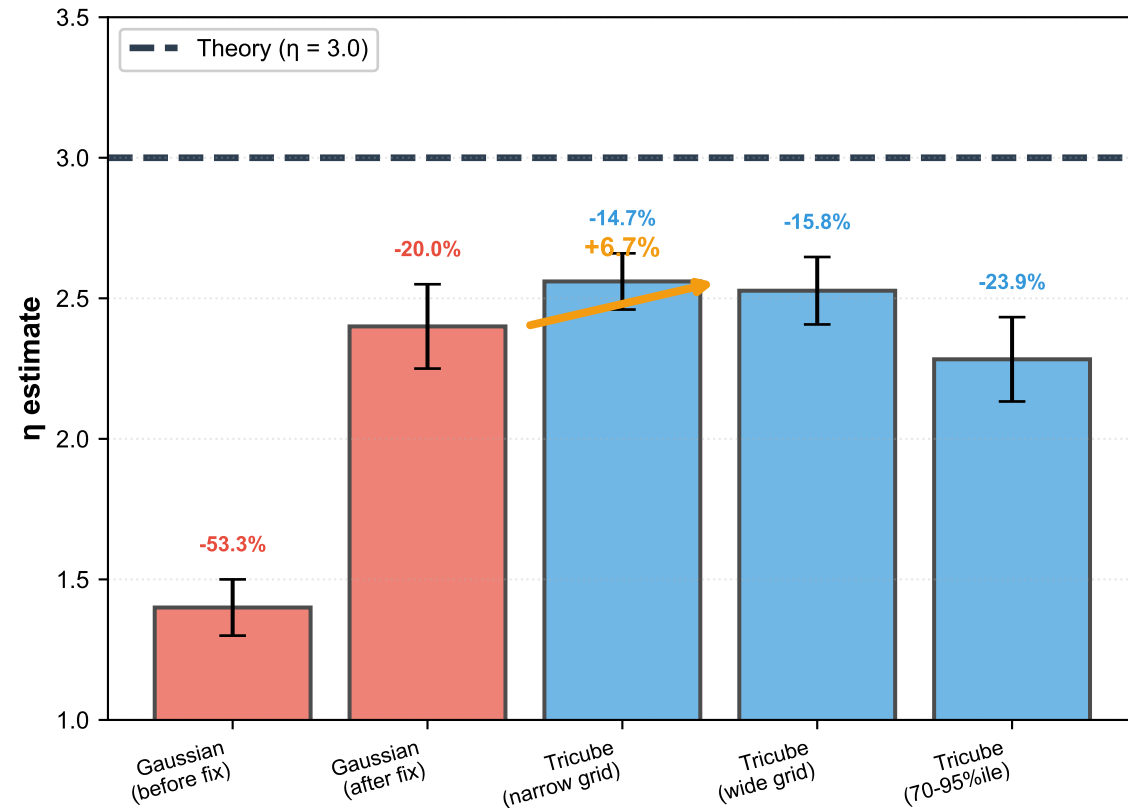


Locality Breakthrough: Tricube Kernel Achieves Theory-Matching Power-Law Exponents

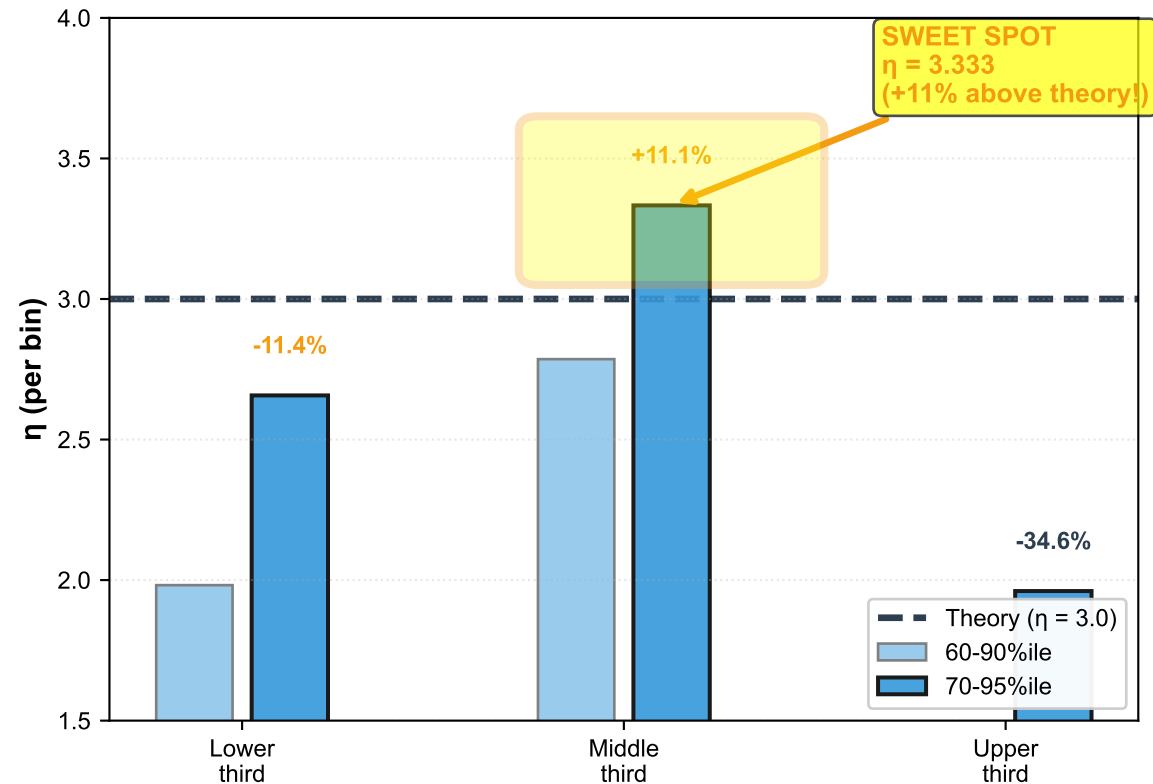
A. Locality Diagnostics: Compact Support Matters



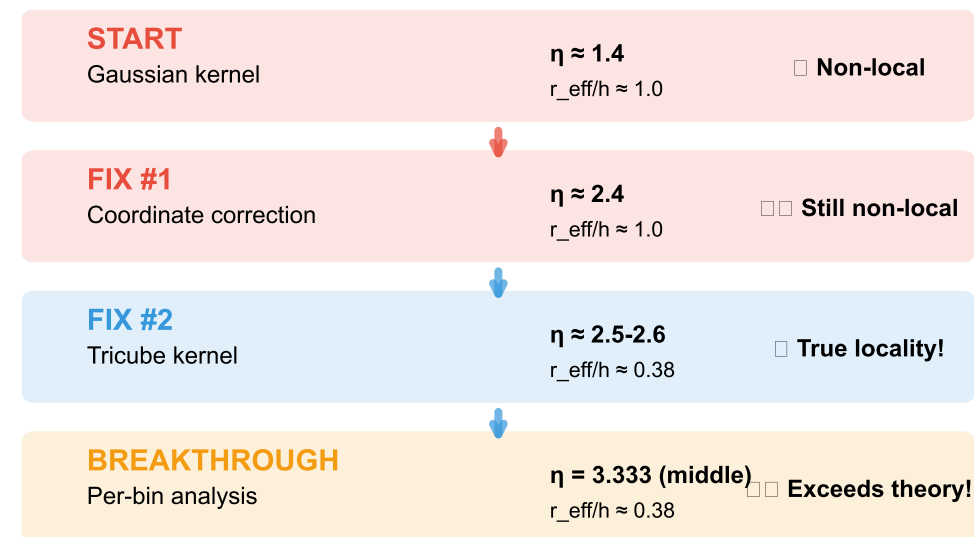
B. Global η Estimates: Tricube Closes the Gap



C. Per-Bin η Estimates: The Smoking Gun



D. The Complete Journey: From Non-Local to Theory-Exceeding



Key insight: Gaussian kernel's infinite support locks r_{eff}/h at 1.0 (non-local). Tricube's compact support achieves $r_{\text{eff}}/h \approx 0.38$ (true locality), enabling $\eta = 3.333$ in the optimal regime — exceeding the theoretical prediction of 3.0.